



VITAMINS DEFICIENCIES & ORAL MANIFESTATIONS

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VITAMINS:

Vitamins are chemically unrelated organic compounds that cannot be synthesized in adequate quantities by humans and therefore, must be supplied by the diet.

WATER SOLUBLE	FAT SOLUBLE
Folic acid	Vitamin A (Retinol, β -carotene)
Cobalamin	Vitamin D (Cholecalciferol)
Ascorbic acid	Vitamin E (Tocopherols)
Pyridoxine	Vitamin K (Phylloquinones, menaquinones)
Thiamine	
Niacin	
Riboflavin	
Biotin	
Pantothenic acid	



DEFICIENCIES & ORAL MANIFESTATIONS

FOLIC ACID (VITAMIN B9):

◆ CLINICAL DEFICIENCY:

MEGALOBLASTIC ANEMIA: Characterized by large, immature red blood cell precursors (megaloblasts) in the bone marrow & macrocytes in the peripheral blood. Macrocytic / Megaloblastic Anemia (MCV > 100 fL) (Normal Range: Generally 80–100 fL.)

NEURAL TUBE DEFECTS: Maternal deficiency during early pregnancy increases the risk of spina bifida & anencephaly.

◆ ORAL MANIFESTATIONS:

HUNTER'S GLOSSITIS: A painful, smooth, and "fiery red" tongue caused by the atrophy of filiform and fungiform papillae.

Recurrent Aphthous Stomatitis

Mucosal Pallor

Angular Cheilitis

Are You Low on Folic Acid? Key Symptoms & What They Mean



FOLIC ACID DEFICIENCY CAUSES



Low Intake

- DIET
- AGE
- ALCOHOL



Decreased Absorption

- CELIAC DISEASE
- CROHN'S DISEASE
- TROPICAL SPRUE
- AMYLOIDOSIS
- SHORT BOWL SYNDROME
- GASTRIC BYPASS



Increased Demand

- PREGNANCY
- INFANCY
- HAEMOLYSIS
- MALIGNANCY
- SEVERE DERMATITIS



Drugs

- METHOTREXATE
- TRIMETHOPRIM
- SULFASALAZINE
- PHENYTOIN
- OCPs

COBALAMIN (VITAMIN B12)

♦ CLINICAL DEFICIENCY :

PERNICIOUS ANEMIA: Often caused by a lack of intrinsic factor (IF), preventing B12 absorption in the ileum. Macrocytic / Megaloblastic Anemia (MCV > 100 fL)

SUBACUTE COMBINED DEGENERATION: Irreversible neurological damage due to failure in myelin maintenance.

♦ ORAL MANIFESTATIONS :

ERYTHEMATOUS GLOSSITIS: Tongue appears “beefy red”, smooth and shiny due to total atrophy of the papillae(often the first clinical sign of deficiency).

BURNING MOUTH SYNDROME (GLOSSODYNIA): A persistent burning sensation of the tongue and oral mucosa without visible lesions.

ORAL PARESTHESIA: Numbness or a “pins and needles” sensation in the tongue and lips reflecting neurological involvement.

DETACHMENT OF EPITHELIUM: Increase mucosal thinning & ulcerations.

Symptoms of Vitamin B12 or Folate Deficiency

SYMPTOMS OF FOLATE DEFICIENCY

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CHRONIC LOW ENERGY



POOR IMMUNE HEALTH
SICK OFTEN



POOR DIGESTION
AND BLOAT



ANEMIA



DEVELOPMENTAL ISSUES IN
PREGNANCY & INFANCY



IRRITABLE
MOOD



OTHER SYMPTOMS:

- PALE SKIN
- CANKER SORES IN MOUTH

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FOLIC ACID DEFICIENCY



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PYRIDOXINE(VITAMIN B6):

♦ CLINICAL DEFICIENCY & TOXICITY:

ISONIAZID INTERACTION: The drug used to treat tuberculosis can induce a B6 deficiency by forming an inactive complex with pyridoxal phosphate (PLP)

SIDEROBLASTIC ANEMIA: Due to impaired heme synthesis. Normal Range: Generally 80–100 fL. Microcytic Anemia (MCV < 80 fL)

TOXICITY: Unlike other water-soluble vitamins, high doses of pyridoxine can cause sensory neuropathy.

♦ ORAL MANIFESTATIONS:

Glossitis

Angular Cheilitis

Stomatitis

Burning Mouth

SYMPTOMS OF VITAMIN B6 DEFICIENCY



SORE GLOSSY
TONGUE



CRACKED AND
SORE LIPS



SKIN RASHES

THIAMINE (VITAMIN B1)

♦ CLINICAL DEFICIENCY:

BERIBERI: This severe thiamine-deficiency syndrome is found in areas where polished rice is the major component of the diet.

DRY BERIBERI: Peripheral neuropathy and muscle wasting.

WET BERIBERI: Edema & cardiovascular failure (high output cardiac failure)

WERNICKE-KORSAKOFF SYNDROME: Frequently seen in chronic alcoholism; involves mental confusion, ataxia, and memory loss. Caused due

♦ ORAL MANIFESTATIONS:

Hyperesthesia

Burning Mouth

Taste Alterations

Vesicular Lesions (though less common, some patients may develop small, herpetiform like vesicles on palate or tongue)

PHYSICAL SYMPTOMS OF VITAMIN B1 DEFICIENCY



FATIGUE AND WEAKNESS



PERIPHERAL NEUROPATHY



HEART ISSUES



OPHTHALMOPELGIA



LOSS OF APPETITE



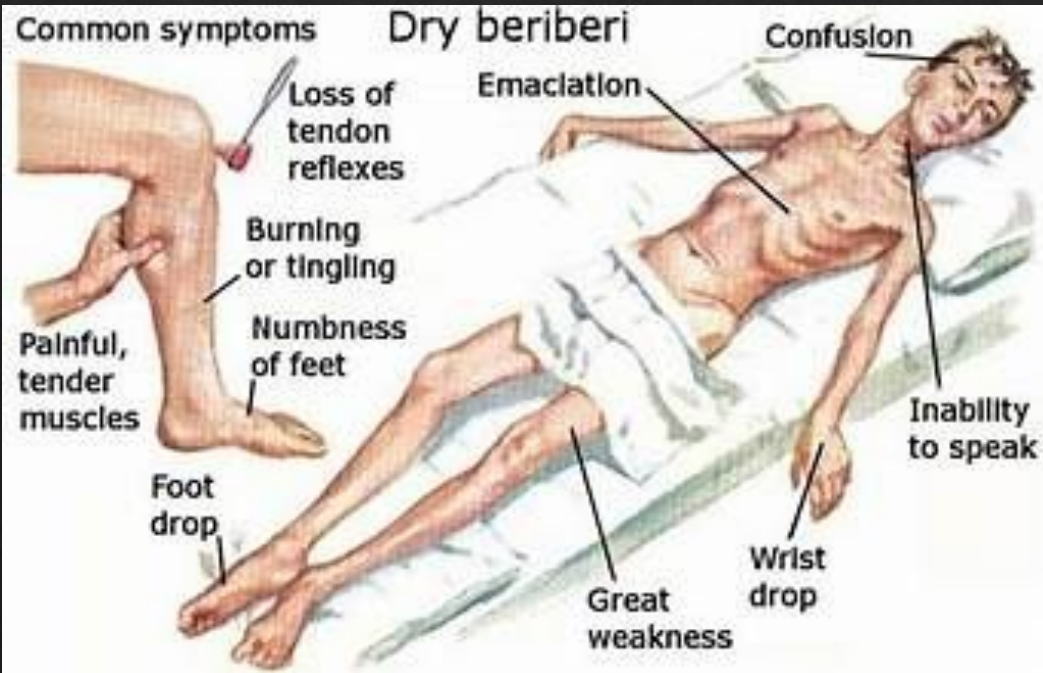
GASTROINTESTINAL SYMPTOMS



BERIBERI



MUSCLE CRAMPS



Substituted pyrimidine

Substituted thiazole

Methylene bridge

CN1C=NC2=C1N=CN2CSC3=CC=CC=C3

Vitamin B1 (Thiamine)

Wet Beriberi

Heart failure and cardiac problems

Swelling of legs

Shortness of breath

Dry Beriberi

Peripheral neuritis

Severe leg pain

paralysis

Food sources of Vitamin B1 (Thiamine):

- Cereals
- Fruits
- Vegetables
- Meat
- Eggs
- Seeds
- Nuts
- Beans

NIACIN (VITAMIN B3)

◆ CLINICAL DEFICIENCY:

PELLAGRA:

The four D's:

1. **Dermatitis** : Sun-sensitive, scaly rash (Casal's necklace)
2. **Diarrhea** : Due to atrophy of the intestinal mucosa.
3. **Dementia** : Neurological symptoms including insomnia & confusion.
4. **Death** : If left untreated.



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♦ ORAL MANIFESTATIONS:

Glossitis (Beefy Red Tongue): The tongue becomes bright red, swollen, and extremely painful.

Angular Cheilitis: Painful inflammation & cracking at the labial commissures (corners of the mouth).

Ulcerative Gingivitis: The gingiva(gums) may become inflamed, bleed easily, and develop painful ulcerations.

Secondary Infections: Patients often suffer from secondary oral infections, such as Vincent's Infection (Trench Mouth), due to weakened mucosal defenses.

NOTE: Niacin is also used as a treatment for hyperlipidemia (inhibiting lipolysis in adipose tissue).

RIBOFLAVIN(VITAMIN B2)

♦ CLINICAL DEFICIENCY:

ARIBOFLAVINOSIS:

General Signs : Dermatitis & corneal neovascularization (blood vessels growing into the cronea).

Etiology : Usually occurs in cases of chronic alcoholism or in combination with other B-vitamin deficiencies.

♦ ORAL MANIFESTATIONS :

Angular Cheilitis: This is the hallmark of B2 deficiency. It presents as painful erythema, scaling, and deep fissures at the corners of the mouth.

Glossitis (Magenta Tongue): The tongue takes on a characteristic purplish-red or magenta hue. The fungiform papillae may become flattened or mushroom-shaped.

Stomatitis: Inflammation of the entire oral mucosa, which may appear thin and pale in some areas while red and inflamed in others.

Seborrheic Dermatitis: Often affects the nasolabial folds and the skin around the lips.



stomatitis of the mouth
and tongue



iron

anemia



migraine
headaches



bloodshot eyes

vitamin
B₂
deficiency



iStock
Credit: Olykava

chronic diarrhea



sore throat



dry skin



cracked red lips

Riboflavin



DAILY NORM

1.8 MG

B₂ + B₃ + B₉ + Zn



Products

Symptoms of deficiency



LOSS OF APPETITE



DERMATITIS



MUSCLE WEAKNESS



ANEMIA



FREQUENT HEARTBURN



SORE MOUTH CORNERS



PHOTOPHOBIA (SENSITIVITY TO LIGHT)



TASTE LOSS



FATIGUE



MUSHROOM
6.4mg(140g)



COTTAGE CHEESE
6.3mg(140g)



LIVER
3.3mg(140g)



ALMOND
6.8mg(140g)



BEEF
6.20mg(140g)



MILK CHOCOLATE
6.45mg(140g)



EGG
6.3mg(140g)

BIOTIN (VITAMIN B7):

◆ Clinical Deficiency:

- Occurrence: Rare, but can be induced by:
 - **Raw Egg Whites**: Contain **Avidin**, a protein that binds biotin tightly and prevents absorption.
 - **Prolonged Antibiotic Therapy**: Depletion of biotin-producing gut microflora.
- Symptoms: Dermatitis, alopecia (hair loss), and loss of appetite.

◆ Oral Manifestations:

- **Glossitis**: Inflammation and soreness of the tongue, often accompanied by a pale, grayish color.
- **Atrophy of Papillae**: The surface of the tongue may become smooth due to the loss of lingual papillae.
- **Angular Cheilitis**: Similar to other B-complex deficiencies, cracking and inflammation may occur at the labial commissures.
- **Localized Paresthesia**: Patients may report tingling or unusual sensations around the mouth and perioral skin.

Deficiency Symptoms of

VITAMIN B7

(biotin)



hair fall



Joint & muscle
pain



brittle nail



hallucination



nausea



dermatitis

FOODS RICH IN

Vitamin b7



carrot



almond



Tomato



onion



eggs



Salmon

Vitamin B7 Deficiency



PANTOTHENIC ACID (VITAMIN B5)

◆ Clinical Deficiency:

- Occurrence: Virtually non-existent in humans under normal conditions.
- Severe Deficiency: "Burning foot syndrome" (paresthesia in the lower extremities) and gastrointestinal distress.

◆ Oral Manifestations:

- Delayed Wound Healing: Because B5 is essential for the synthesis of fatty acids and membrane phospholipids, a deficiency can impair the repair of the oral mucosa after trauma or surgery.
- Glossitis: General inflammation and redness of the tongue.
- Stomatitis: Inflammation of the mucous lining of the mouth, which may lead to painful ulcerations.
- Angular Cheilitis: Fissuring and cracking at the corners of the lips, often seen when B5 deficiency occurs alongside other B-complex deficits.



ASCORBIC ACID (VITAMIN C)

◆ Clinical Deficiency: Scurvy

- Etiology: Diet lacking fresh fruits and vegetables.
- Pathophysiology: Failure of collagen to cross-link effectively, leading to decreased tensile strength of connective tissues and capillary fragility.

◆ Oral Manifestations:

- Gingival Changes: Gums become swollen, spongy, and take on a "bluish-red" or "purple" hue.
- Hemorrhage: Spontaneous bleeding of the gingiva, even with minimal provocation.
- Tooth Mobility: Severe weakening of the **Periodontal Ligament (PDL)**, leading to the loosening and eventual loss of teeth.
- Impaired Healing: Significant delay in the healing of extraction sockets or mucosal ulcers.



Figure 2 : Corkscrew hair [3]



Figure3: gingivitis



Vitamin C Deficiency

Deficiency disease: **Scurvy**

Deficiency symptoms:

- **S**kin discoloration & bruising
- **H**emorrhaging
- **A**naemia
- **D**ental issues
- **E**xhaustion / fatigue
- **S**welling of joints (edema)

Mnemonic: **SHADES**



Normal Person



Healthy teeth
and gums



Scurvy

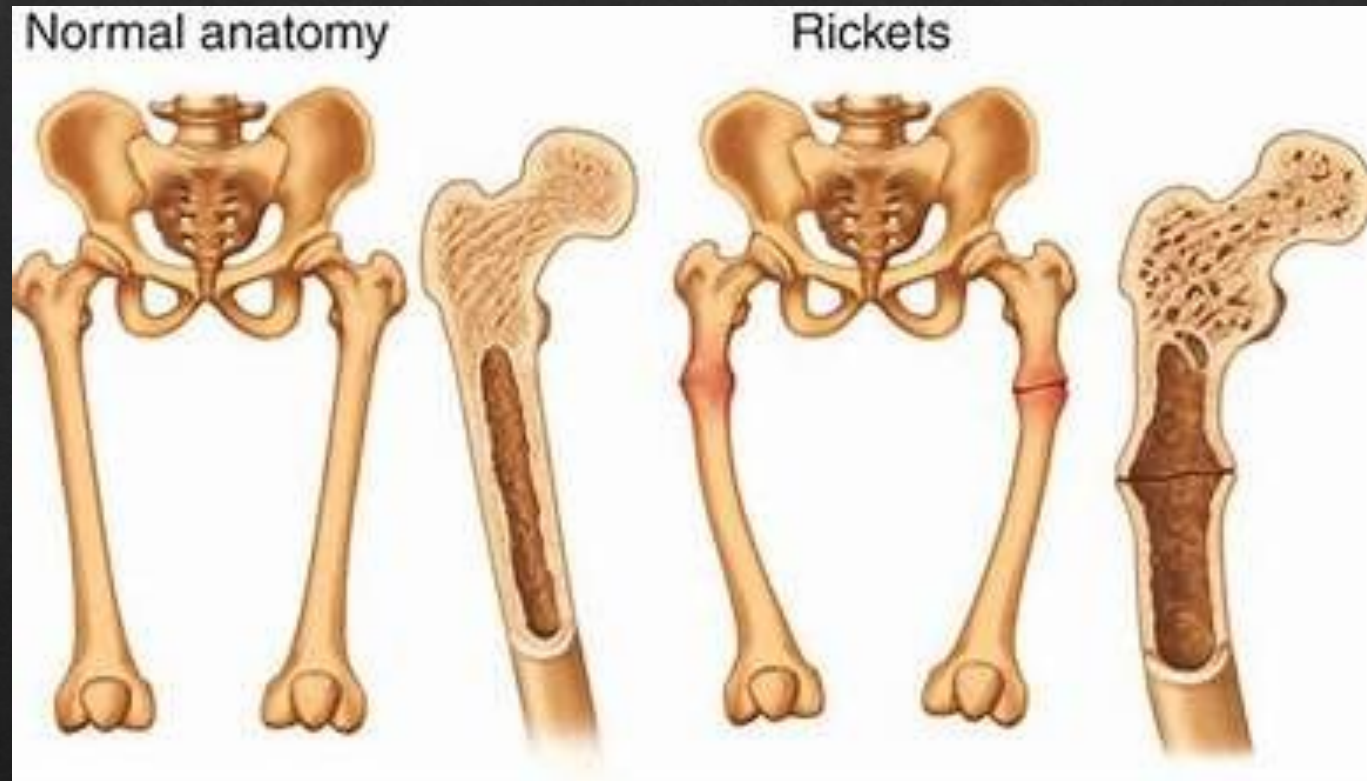
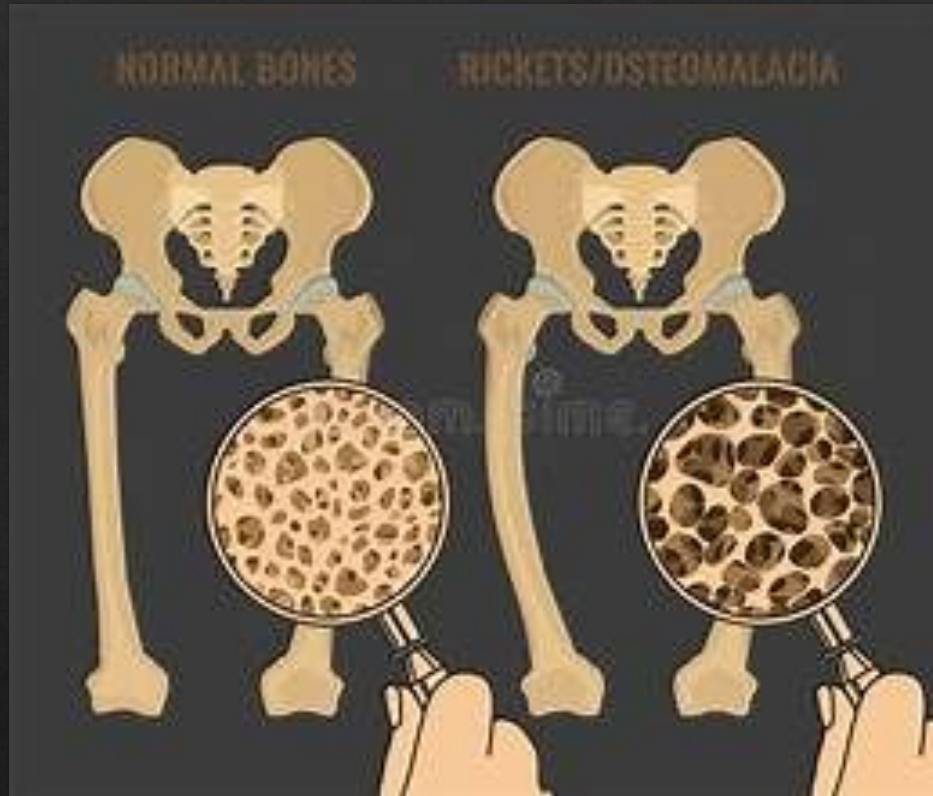


Loosened teeth and
bleeding / ulcerated gums

VITAMIN D

- ◆ Clinical Deficiency:

- Rickets (Children): Continued formation of collagen matrix but incomplete mineralization, leading to "soft bones" and skeletal deformities.
- Osteomalacia (Adults): Demineralization of pre-existing bone, increasing the risk of fractures.



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◆ Oral Manifestations:

- Enamel Hypoplasia: If deficiency occurs during tooth development, it results in permanent defects in the enamel (pitting or thinning).
- Delayed Eruption: Significant delay in the eruption of both primary and permanent teeth.
- Alveolar Bone Loss: Weakening of the bone supporting the teeth, which can exacerbate periodontal disease.
- Dentine Defects: Large pulp chambers and "interglobular dentin" (unmineralized zones within the dentin matrix).
- Malocclusion: Due to the softening of the mandible and maxilla, the jawbones may deform under the pressure of mastication.

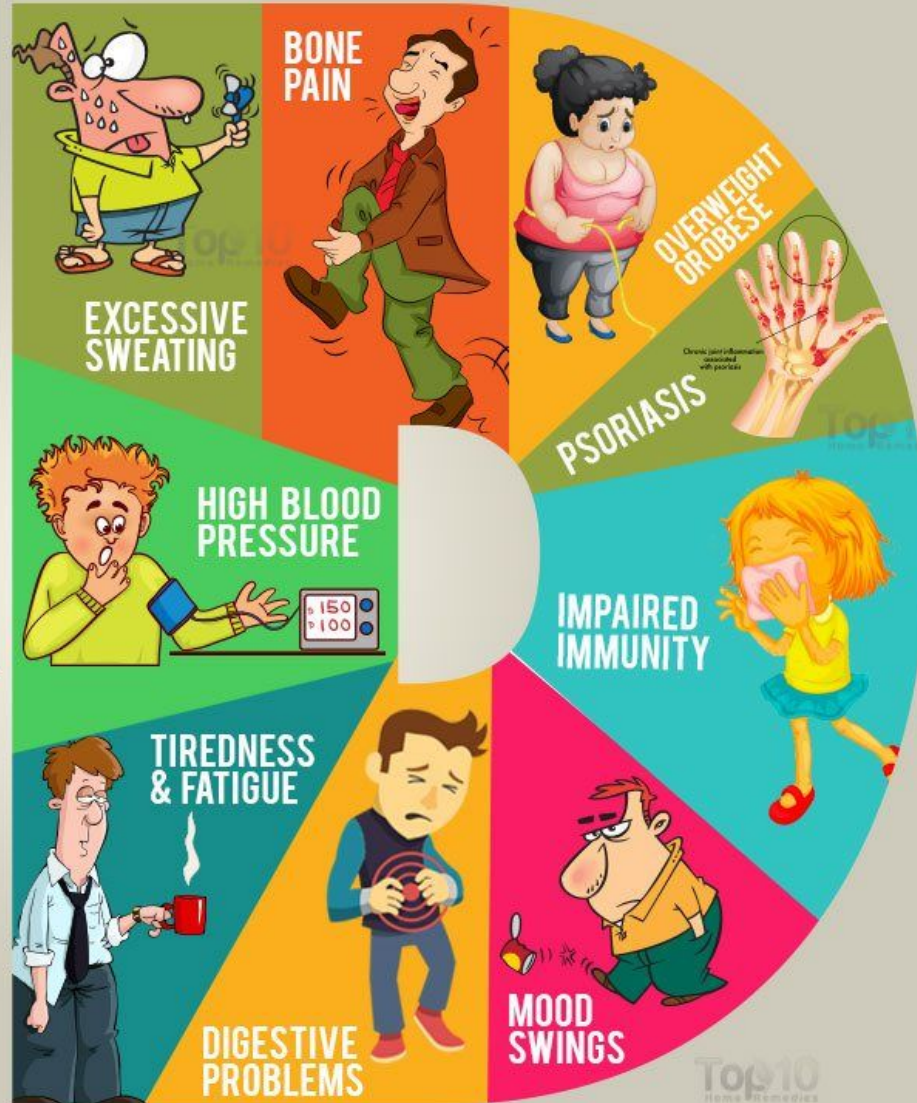
SIGNS

You May Have a

VITAMIN D DEFICIENCY

Top10
Home Remedies

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thank you!

